

BIOLOGY GRADE 10: CHEMICALS OF LIFE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.2 (6 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 1.0: Cell Biology and Biodiversity
Sub-Strand	Sub-Strand 1.2: Chemicals of Life
Total Duration	6 lessons × 40 minutes = 240 minutes total
Sub-Strand Content	– Biological molecules: carbohydrates (monosaccharides, disaccharides, polysaccharides) – Lipids: fats, oils and waxes – Proteins: amino acids and polypeptides – Food tests: starch (iodine), reducing sugars (Benedict's), proteins (Biuret), lipids (emulsion/ethanol and brown paper test) – Roles of carbohydrates, lipids and proteins in living organisms – Water and mineral salts as chemicals of life – Enzymes: nature, properties and factors affecting enzyme action (temperature, pH)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - describe the chemical composition of carbohydrates, lipids and proteins found in living organisms, - carry out food tests to identify starch, reducing sugars, proteins and lipids in food samples, - explain the roles of carbohydrates, lipids and proteins in living organisms, - investigate the effect of temperature and pH on enzyme action, - appreciate the importance of the chemicals of life in everyday nutrition and health.
Core Competencies	– Critical Thinking and Problem Solving: Learners analyse food test results and make evidence-based claims about the nutrient content of Kenyan foods. – Communication and Collaboration: Learners work in groups to carry out food tests, share findings and revise a class model. – Learning to Learn: Learners track their growing understanding using the Driving Question Board and reflect on how their ideas have changed. – Digital Literacy: Learners use simulations and digital resources to visualise molecular structures.
Core Values	– Responsibility: Learners handle chemicals and food samples safely and dispose of waste properly. – Unity: Learners cooperate in groups when carrying out experiments. – Integrity: Learners record and report experimental results honestly. – Respect: Learners value diverse ideas during discussions.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate questions for the Driving Question Board about why ugali and githeri sustain energy.

	– Developing and Using Models: Learners build and revise a model linking chemicals in food to body function. – Planning and Carrying Out Investigations: Learners design and conduct food tests and enzyme experiments. – Analyzing and Interpreting Data: Learners interpret colour changes and enzyme activity data. – Constructing Explanations: Learners explain nutrient roles using evidence. – Engaging in Argument from Evidence: Learners defend claims about the best food for athletes.
Pertinent & Contemporary Issues (PCIs)	– Health Education: Learners relate balanced diets to nutrition-related conditions such as malnutrition and kwashiorkor. – Safety and Security: Learners follow laboratory safety rules when using iodine, Benedict's and Biuret reagents. – Environmental Conservation: Learners use locally available food materials and dispose of waste responsibly.
Career Connections	– Nutritionist/Dietitian: Plans balanced diets using knowledge of carbohydrates, lipids and proteins. – Biochemist: Studies the chemical molecules of living organisms. – Food Technologist: Tests and processes foods such as flour and cooking oil at KEBS-regulated factories. – Medical Laboratory Technologist: Performs biochemical tests on samples. – Agricultural Extension Officer: Advises farmers on nutritious crops and livestock feeds.
Focus for Lessons	This unit helps learners discover that the energy, growth and strength seen in Kenyan athletes come from the chemicals in everyday foods. Learners investigate through food tests, enzyme experiments and modelling. They revise a cumulative model throughout the six lessons to connect food molecules to body function.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do the simple foods Kenyan athletes eat, like ugali, githeri and beans, provide the chemicals their bodies need for energy, growth and repair? KICD KEY INQUIRY QUESTIONS: 1. What are the chemicals of life? 2. How do chemicals of life contribute to the functioning of living organisms?
Anchoring Phenomenon	Kenyan long-distance runners from Iten in the Rift Valley, such as Eliud Kipchoge, traditionally eat a diet dominated by ugali, githeri, sukuma wiki, beans and milk, with little processed food. Yet they run at world-record pace for hours. Learners observe that a plate of ugali and beans looks plain and unrelated to the athlete's performance. The puzzle is how simple foods supply the energy, repair and body-building materials for such extreme performance, and what chemicals in the food make this possible.
Storyline Thread	Lesson 1: Predict what makes up a plate of ugali and beans and why athletes eat it; introduce the phenomenon, create the Driving Question Board and draw an initial model; introduce carbohydrates, lipids and proteins as chemicals of life. Lesson 2: Carbohydrates, structure (monosaccharides, disaccharides, polysaccharides) and tests for starch and reducing sugars on maize flour, sugarcane juice, ripe bananas and sweet potato. Lesson 3: Proteins and amino acids; Biuret test on beans, milk, eggs and omena; explain the roles of proteins in growth and repair; add to DQB and revise the model. Lesson 4: Lipids: fats, oils and waxes; emulsion and brown paper tests on groundnuts, cooking oil, mandazi and nyama choma; explain the roles of lipids in energy storage and insulation. Lesson 5: Enzymes: why saliva makes ugali taste sweet; investigate the effect of temperature and pH on amylase or catalase activity; interpret data and explain enzyme specificity. Lesson 6: Synthesis: analyse and compare nutrient content of Kenyan meals, argue from evidence which meal best supports an athlete, finalise the cumulative model and answer the Driving Question Board.
Supporting Phenomena	– Chewing a piece of ugali for a long time makes it taste sweet, showing starch being broken down by enzymes in saliva.

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| | <ul style="list-style-type: none">– A mandazi or a piece of nyama choma leaves a translucent grease spot on paper, showing the presence of lipids.– Milk left out in Kisumu's heat curdles into mala or sours, showing how temperature and pH affect proteins.– Pawpaw (papain) tenderises tough meat in Kenyan cooking, showing enzyme action on proteins. |
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (40 minutes (one double or two single periods may be used if the class is large)): A Plain Plate with Extraordinary Results

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Launch the anchoring phenomenon: a plain plate of ugali, beans, sukuma wiki and milk fuels Kenyan marathon runners from Iten. Learners discover that this food is made of chemicals of life, meet the three main groups (carbohydrates, lipids and proteins) with a one-line role each, draw an initial model and start the Driving Question Board.
Knowledge	Learners name carbohydrates, lipids and proteins as the three main groups of chemicals of life and state one role for each: carbohydrates supply and store energy, proteins build and repair the body, lipids store a lot of energy and insulate. They recognise that water, vitamins, minerals and nucleic acids are also chemicals of life, to be met later. They note that a marathon costs about 2,200 kcal and stored glycogen is about 2,000 kcal.
Skills	Making predictions from prior knowledge; analysing simple data (pace, Table 1 values); asking questions; developing and using an initial model; communicating ideas in groups of 4. NGSS practices: Asking Questions, Developing and Using Models, Analysing and Interpreting Data.
Attitudes	Curiosity about everyday Kenyan foods; respect for each group member's ideas; appreciation that familiar foods such as ugali contain chemicals that support extraordinary performance; willingness to revise ideas when new evidence arrives.
Key Inquiry Question	1. What are the chemicals of life? 2. How do chemicals of life contribute to the functioning of living organisms?
Purpose in Storyline	This lesson sets up the whole unit. Learners meet the puzzle of how simple foods fuel world-record running. Their predictions, DQB questions and initial model are the starting point that Lessons 2 to 6 will test and revise. Lesson 2 will study carbohydrates, Lesson 3 proteins, Lesson 4 lipids, Lesson 5 enzymes, and Lesson 6 will use Table 4 meals to give the final explanation.
Safety Notes	No chemical tests or tasting of lab samples in this lesson. If a teacher brings real food for display (ugali, beans, sukuma wiki, milk), keep it for looking only, check for allergies (milk, groundnuts) and keep hands clean. Use chart paper and markers safely and keep walkways clear when learners move to the DQB.

B. LESSON OVERVIEW

Learners begin by looking at a picture or real display of a plate of ugali, beans, sukuma wiki and milk, the everyday food of athletes in Iten, Rift Valley. They predict individually and then in groups of 4 what the plate is made of and why athletes such as Eliud Kipchoge eat it. In the Observe phase they look at Kipchoge's marathon figures (7,269 s over 42.195 km, 172.3 s per km, about 2,200 kcal needed, glycogen about 2,000 kcal) and a brief look at Table 1 (nutrients per 100 g). Learners notice that the food holds carbohydrate, protein and fat, and meet each of the three groups with a one-line role. In the Explain phase they use a sentence frame to link the plain plate to the performance. They then write questions on sticky notes to build the class Driving Question Board, and finally draw an initial model on paper, which is merged into one class model on chart paper that will be revised in every lesson.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Introduction to carbohydrates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Carbohydrates (Flexbook) Source: CK-12  Search ARES for similar readings	Learners look at a picture or real display of a plate of ugali, beans, sukuma wiki and a cup of milk. The teacher reads the phenomenon: Kenyan runners from Iten, such as Eliud Kipchoge, eat mainly ugali, githeri, sukuma wiki, beans and milk with little processed food, yet run at world-record pace for hours. Individually (2 minutes) learners write answers to: (1) What is this plate made of? (2) Why do you think athletes eat it? (3) How might it help someone run a marathon? They then share in groups of 4, agreeing on one prediction to write on a group card. They use only ideas from home and primary school.	Display the plate and say: "This looks plain. Nothing fancy, nothing from a packet. Yet this is what many champions from Iten eat." Ask: "What do you think is inside this food that could power someone for two hours?" Give WAIT TIME of 10-15 seconds before anyone speaks. Say: "Write first, talk second." Circulate and listen for words such as energy, strength, vitamins, protein, sugar. Cold-call 4 groups (not volunteers) to read their group prediction aloud. Say: "I am not looking for the right answer today. I am looking for your honest thinking." Record key words on the board without judging them. Ask: "Who has a different idea from this group?" and take 2 more responses.	Eliciting prior knowledge through Predict-Share-Compare (Think-Pair-Share extended to groups of 4). Writing before talking commits each learner to an idea, so later evidence can be compared with the starting prediction and learners notice how their thinking changes.	Look for: (1) every learner has a written prediction before group talk (check at least 10 papers while circulating); (2) predictions use food or body words such as energy, strength, protein, sugar or vitamins; (3) groups can give a reason, not only a word (for example, 'ugali gives energy because it is heavy and fills you'); (4) note misconceptions such as 'ugali is just filler' or 'only meat gives strength' to revisit in later lessons.
Observe Phase  VIDEO: Introduction to carbohydrates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Carbohydrates (Flexbook) Source: CK-12  Search ARES for similar readings	Learners study two evidence sets in their groups of 4. Set 1: Kipchoge data card. Marathon world record 2:01:09 at Berlin 2022 = 7,269 s over 42.195 km. Learners calculate average pace: $7,269 / 42.195 = 172.3$ s per km, about 2 min 52 s per km, about 20.9 km/h. Body mass about 52 kg. A marathon costs about 1 kcal per kg per km, so about $52 \times 42.195 = 2,194$, so about 2,200 kcal. Stored glycogen in muscle and liver is about 500 g = about 2,000 kcal. Set 2: a brief look at Table 1 (per 100 g): ugali 26 g carbohydrate / 2.5 g protein / 0.6 g fat = 119.4 kcal; githeri $18 / 6 / 1.0 = 105$ kcal; boiled beans $21 / 8 / 0.5 = 120.5$ kcal; sukuma wiki $5 / 2.5 / 3 = 57$ kcal; whole milk $4.7 / 3.3 / 3.9 = 67.1$ kcal. Learners circle the biggest number in each row and write what they notice and wonder. The teacher then names	Hand out the data card and say: "Before you calculate, look at the numbers. What stands out?" WAIT TIME of 10-15 seconds. Guide the pace calculation: "Divide 7,269 seconds by 42.195 kilometres. What do you get?" Check that groups reach 172.3 s per km, then ask: "Could you run even one kilometre that fast? Think of the fastest kilometre you have run at school." Ask: "About how much energy does the marathon cost, and about how much can the body store as glycogen?" Cold-call 3 groups for the 2,200 kcal and 2,000 kcal figures. Say: "Notice how close those two numbers are. Keep that in mind." Move to Table 1: "Look at ugali. Which column is biggest?" Cold-call 3 groups. Do NOT explain structure or tests. Say: "Scientists call these three groups chemicals of life: carbohydrates, proteins and lipids.	Noticing and Wondering with data. Learners first notice patterns in numbers (carbohydrate is the largest column for ugali, githeri and beans) before the teacher gives the vocabulary, so the names carbohydrates, proteins and lipids attach to evidence they have already seen. Simple calculations make the extreme performance concrete.	Look for: (1) groups calculate 172.3 s per km correctly (or within rounding) from $7,269 / 42.195$; (2) learners identify carbohydrate as the largest value in ugali, githeri and beans in Table 1; (3) learners state that the marathon needs about 2,200 kcal and that glycogen holds about 2,000 kcal; (4) in the 3 cold-call responses, check that learners match each group name to its one-line role; (5) record 'I notice / I wonder' statements that will become DQB questions.

	the three groups in one line each: carbohydrates supply and store energy; proteins build and repair the body; lipids store a lot of energy and insulate. Water, vitamins, minerals and nucleic acids are named as other chemicals of life, to be met later.	Carbohydrates supply and store energy. Proteins build and repair the body. Lipids store a lot of energy and insulate." Write the three names with their one-line roles on the board. Add: "Water, vitamins, minerals and nucleic acids are also chemicals of life. We will not study them yet." Ask: "Where in this plate do you think each group is hiding?" and take 3 ideas after WAIT TIME.		
Explain Phase  VIDEO: Matter and energy in food webs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Carbohydrate Classification (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Using their predictions, the data card and the three named groups, learners write a first explanation in their notebooks with this sentence frame: 'A plate of ugali and beans is not plain because it contains _____. For example, _____ supplies _____ and _____ helps to _____. A marathon needs about _____ kcal, and the body stores about _____ kcal as glycogen, so the athlete also needs _____.' They compare it with their earlier prediction, underline what has changed in one colour, and share with a partner. Two or three learners read out their explanations to the class. The class lists what they still cannot explain, such as how food becomes energy in the muscle.	Say: "You started this lesson with a prediction. Now use evidence to improve it." Display the frame. Give WAIT TIME of 10-15 seconds, then 4 minutes writing. Circulate and ask: "Which number from the data card supports your sentence?" and "Which group of chemicals does ugali mostly give?" Prompt struggling learners: "Look at the largest value in the ugali row." Cold-call 3 learners to read their explanations. After each ask the class: "What evidence did they use?" Then ask: "The marathon costs about 2,200 kcal and glycogen holds about 2,000 kcal. What does that tell you about why eating matters on race day and in training?" Take 2 responses after WAIT TIME. Say: "Good scientists also say what they do not know yet. What can't we explain?" Record 3 gaps on the board. Do not answer them; say: "We will return to these."	Claim-Evidence-Reasoning scaffold through a sentence frame. Learners connect a claim (the plate is not plain) to evidence (Table 1 and Kipchoge figures) and reasoning (chemical groups have roles). Comparing with the earlier prediction makes the change in their thinking visible and builds scientific habits of revision.	Look for: (1) the explanation names at least two of carbohydrates, proteins, lipids with a correct role; (2) at least one number from the data card or Table 1 is used as evidence; (3) learners link the 2,200 kcal need to food and the 2,000 kcal glycogen store without calculating further; (4) learners mark what changed from their prediction; (5) common errors to note: calling ugali a protein food, saying fat is always bad, or confusing energy with vitamins.
Driving Question Board (DQB) Creation  VIDEO: Food and energy in organisms <i>Related topic: not</i>	The teacher displays the Driving Question: 'How do the simple foods Kenyan athletes eat, like ugali, githeri and beans, provide the chemicals their bodies need for energy, growth and repair?' on a large chart at the front. Each learner writes 2 questions on	Unveil the chart and say: "This is our Driving Question. By the end of this unit, we will answer it with evidence." Say: "Write two questions you really want answered. Questions beginning with how or why are strong." WAIT TIME of 10-15 seconds before	Driving Question Board as a tool for student-generated questions. Learners ask questions based on their own noticing, which gives them ownership of the investigation. Clustering shows how questions connect to later lessons and gives a purpose for	Look for: (1) every learner has written at least 2 questions (count notes per group); (2) at least half the questions are testable or explainable, not just yes/no; (3) questions refer to the phenomenon (ugali, beans, Kipchoge, energy) rather than random topics; (4)

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Carbohydrate Classification Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	sticky notes, using their wonderings and the class gap list (for example, 'How do we know ugali has starch?', 'What are proteins made of?', 'Why do athletes need fat?', 'How does the body break food down?'). In groups of 4 they pick their best question, place it on the DQB and explain it in one sentence. As a class they sort the notes into clusters (for example: what is in food, how to test food, what proteins do, how digestion works, which meal is best).	writing starts. Circulate and coach: "Can you make that question more specific to ugali or to Kipchoge?" Cold-call 5 groups to place one note each and read it aloud. Ask: "Which questions seem to belong together?" and guide the clustering. Point to the two KICD questions and say: "What are the chemicals of life? How do they help living things work? Can you find notes that match these?" Add the missing questions yourself if needed. Say: "We will come back to this board in every lesson, move questions we have answered and add new ones." Photograph the board for records.	each investigation.	groups give a reason when placing notes; (5) at least one cluster matches each KICD key inquiry question.
Model Building Phase VIDEO: Dehydration synthesis or a condensation reaction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Carbohydrates Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Individually, learners draw an initial model on a sheet of paper, answering: 'How does a plate of ugali, beans, sukuma wiki and milk give an athlete energy, growth and repair?' The model should show: the food on the plate, the three groups of chemicals (carbohydrates, proteins, lipids) with arrows, the athlete's body (muscles), and a question mark where they do not yet know what happens. Learners label what each group does using the one-line roles. In groups of 4 they compare models, then each group sends one member to add a feature to the class model on chart paper. The class model is displayed next to the DQB as 'Model Version 1'.	Say: "Draw what you think happens from the plate to the runner. Use arrows and labels. Draw question marks where you are unsure; that is honest science." Give WAIT TIME of 10-15 seconds, then 6 minutes drawing. Circulate and ask: "Where does the energy come from in your picture? Where do repair and growth fit in?" Cold-call 3 groups to explain one feature of their model. Ask the class: "What is similar across the models? What is different?" Build the class model by inviting 4 learners to add one element each. Say: "This is Model Version 1. It will change. After each lesson we will add evidence, fix mistakes and finally explain how simple foods fuel a 2-hour marathon." Close with the template celebration: "You didn't just learn ABOUT science—you DID science!" Ask learners to bring a food label or write down one food they ate yesterday for Lesson 2. Optional homework: ask a family member what they think gives	Developing and revising a cumulative model. An initial model makes learners' ideas visible, including gaps (question marks). Because the same model is revised in each lesson, learners can see their understanding grow and link each new chemical group back to the phenomenon.	Look for: (1) models show the plate, the body and arrows linking them; (2) at least two of carbohydrates, proteins and lipids are labelled with a correct one-line role; (3) question marks or gaps are shown honestly; (4) learners can explain one arrow when asked; (5) misconceptions to note (for example, energy shown as 'coming from vitamins' or 'water is the main energy source') to address in later lessons. Collect a few models to compare with the final model in Lesson 6.

		athletes strength.		
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D. TEACHER REFLECTION

After the lesson, consider: Did every learner write a prediction before talking? Which predictions and misconceptions were most common (for example, 'ugali is only for filling the stomach')? Could groups use the Kipchoge data to see why the food matters (pace of 172.3 s per km, about 2,200 kcal needed, about 2,000 kcal glycogen)? Were the one-line roles for carbohydrates, proteins and lipids clear without going into structure or tests? Did the DQB questions reflect the whole range of ideas, and do they cover both KICD inquiry questions? Which learners were quiet and need a different way to contribute next time? Did the initial model show honest question marks? Note which DQB clusters to address first in Lesson 2 and how to use the sticky notes in each lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What did we observe? Kipchoge ran 42.195 km in 7,269 s, a pace of 172.3 s per km, and the marathon costs about 2,200 kcal while stored glycogen is about 2,000 kcal. Table 1 shows that ugali, githeri and beans contain carbohydrate, protein and a little fat.
What did I learn?	What did we learn? Carbohydrates, proteins and lipids are the three main groups of chemicals of life. Carbohydrates supply and store energy, proteins build and repair the body, and lipids store a lot of energy and insulate. Water, vitamins, minerals and nucleic acids are also chemicals of life, which we will meet later.
How does this explain the phenomenon?	How does this help explain the phenomenon? A plate of ugali, beans, sukuma wiki and milk is not plain: it contains the chemicals of life that the body needs for energy, growth and repair. We still need to find out how we can show these chemicals are in the food and how the body uses them, so we can explain how simple foods fuel a marathon runner from Iten.

LESSON 2 (80 minutes (double lesson)): Carbohydrates: starch, sugars and how to test for them

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how maize in ugali and other Kenyan foods contain carbohydrates of different structures, and use the iodine and Benedict's tests to show which foods hold starch, reducing sugar or non-reducing sugar, so they can explain how ugali supplies energy to athletes like Eliud Kipchoge.
Knowledge	By the end of the lesson, learners should be able to: (a) classify carbohydrates as monosaccharides (glucose, fructose), disaccharides (sucrose, maltose) and polysaccharides (starch, glycogen, cellulose); (b) state that reducing sugars (glucose, fructose, maltose) give a positive Benedict's test while the non-reducing sugar sucrose does not until it is broken down by boiling with dilute HCl; (c) state that iodine turns blue-black with starch; (d) compare carbohydrate per 100 g in ugali (26 g), ripe banana (23 g), boiled sweet potato (20 g) and sugarcane juice (18 g).
Skills	Planning and carrying out investigations (NGSS SEP 3): carry out iodine and Benedict's tests safely using a hot water bath. Analysing and interpreting data (SEP 4): record colour changes in a table and link them to carbohydrate type. Developing and using models (SEP 2): add carbohydrate structure to the class model. Constructing explanations (SEP 6): use test evidence to explain why ugali is a good energy food.
Attitudes	Learners show curiosity about everyday Kenyan foods, care and responsibility in following safety rules, honesty in recording what they actually see, and respect for group members' ideas and roles.
Key Inquiry Question	What are the chemicals of life? How do chemicals of life contribute to the functioning of living organisms?
Purpose in Storyline	Lesson 1 named carbohydrates as the energy chemical and showed that ugali is made from maize. This lesson gathers the first piece of lab evidence: what kind of carbohydrate is in maize, sweet potato, banana and sugarcane, and how to prove it. It prepares learners to add protein in Lesson 3, and to understand in Lesson 5 why chewed ugali tastes sweet. It also moves learners towards the final comparison of athlete meals in Lesson 6.
Safety Notes	Goggles must be worn for the whole practical. Benedict's solution is heated only in a hot water bath (never over a naked flame), with test tubes pointed away from faces and a test-tube holder used. Dilute HCl is handled only by the teacher or by learners under direct supervision, and is neutralised with sodium hydrogencarbonate before retesting. Iodine stains skin and clothes. NO tasting of any laboratory sample. Wash hands after the practical. Wipe spills immediately.

B. LESSON OVERVIEW

Lesson 1 left learners with a puzzle: a plain plate of ugali and beans seems unrelated to Eliud Kipchoge's 2:01:09 marathon, which costs about 2,200 kcal. Learners named carbohydrates as the energy chemical and drew an initial model. In this lesson they ask: what exactly is the carbohydrate in ugali, and how can we prove it? Learners first predict which foods contain starch or sugar. They then test maize flour, sweet potato, ripe banana and sugarcane juice with iodine and Benedict's solution, with a surprising result: sugarcane juice tastes very sweet but Benedict's stays blue until it is boiled with dilute HCl. They explain the results using the structure of monosaccharides, disaccharides and polysaccharides, and compare carbohydrate per 100 g in Table 1 foods. The DQB and the class model are updated with carbohydrate structure and tests.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Carbohydrates Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(10 min) Recall from Lesson 1: carbohydrates are the energy chemical and ugali is made from maize. Learners see four samples on the teacher's table: dry maize flour, sweet potato, ripe banana and sugarcane juice. Individually, then in groups of 4, they complete a prediction table on a sheet: for each food, predict if it contains starch, sugar, both, or neither, and write one reason. They also answer: 'Sugarcane juice from Kisumu tastes very sweet. Do you think a sugar test will be positive? Why?' Each group draws a quick sketch of what they think a starch molecule looks like compared to a sugar molecule (for example, one bead versus a long chain of beads).	Hold up the ugali picture from Lesson 1 and say: "Last lesson we said ugali carries the energy chemical. Today we ask: which carbohydrate is it, and how can we prove it?" Ask: "Which of these four foods do you think has the most starch? Write your prediction silently." WAIT TIME 10-15 seconds. Then say: "Turn to your group and agree on one prediction for each food." Cold-call 3 groups: "What did you predict for sugarcane juice, and why?" Record predictions on the board in a class prediction table and do NOT say which are right. Say: "Keep your predictions; we will check them with evidence." Circulate and note any ideas such as 'sweet means sugar, so it must test positive' for later use.	Prediction with reasoning (POE: Predict-Observe-Explain). Writing a prediction and a reason before seeing data surfaces learners' ideas, so a surprising result such as sugarcane juice not reacting with Benedict's creates a real cognitive conflict and a need to explain.	Observe whether learners (1) give a reason with each prediction, not only a guess, (2) link 'sweet' with sugar and 'staple food' with starch, and (3) show in their sketch the idea of one unit versus a chain. Tally on the board how many groups predict sugarcane juice will be positive. Learners who write only 'yes' or 'no' without a reason get a prompt: 'What makes you think so?'
Observe Phase  VIDEO: Hydrolysis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Carbohydrate Classification (Flexbook) Source: CK-12  Search ARES for similar readings	(30 min) Groups of 4 with assigned roles (tester, recorder, timekeeper/safety officer, reporter) wearing goggles carry out food tests on four samples. Part A, iodine test: add 2 drops of iodine solution to about 2 cm ³ of each sample (maize flour paste, mashed sweet potato, mashed ripe banana, sugarcane juice) on a white tile or in a test tube and record the colour. Part B, Benedict's test: add about 2 cm ³ of Benedict's solution to about 2 cm ³ of each sample, heat in a hot water bath for 3-5 minutes, and record the colour change (blue to green to orange to brick-red). Expected results in the class table: Maize flour: iodine blue-black, Benedict's stays blue. Sweet	Before starting, hold up goggles and say: "No goggles, no practical. Benedict's is heated only in the hot water bath. Nobody tastes anything in this lab." Demonstrate with one tube how to read the colour scale blue-green-orange-brick-red. Say: "Recorders, write exactly the colour you see, not the colour you expect." Circulate and ask each group: "What colour did the iodine give on maize flour? What does that tell you?" For the HCl step say: "Watch carefully. Sugarcane juice stayed blue. Now I boil it with acid. Predict silently what will happen." WAIT TIME 10-15 seconds before showing the retest. Then ask: "What changed between the first and second test?" Cold-call 3 learners for their	Structured data collection and pattern-finding in a class results table. Pooling results from all groups lets learners see consistent patterns (starch gives blue-black, ripe banana loses starch), and spot the anomaly (sweet sugarcane juice only reacts after acid), which builds evidence-based reasoning rather than memorised facts.	Check that learners (1) wear goggles and use the water bath correctly, (2) record colours exactly (for example 'green-yellow' not just 'positive'), (3) obtain results matching Table 2, and (4) can say which food gave iodine blue-black. Use a quick checklist for each group. Ask the reporter: 'Which two foods gave the same iodine result and which gave different Benedict's results?' Groups whose results differ are asked to repeat the test and explain possible reasons such as too little heating time or too little sample.

	potato: iodine blue-black, Benedict's green-yellow. Ripe banana: iodine faint brown-orange, Benedict's orange to brick-red. Sugarcane juice: Benedict's stays blue. Part C: teacher demonstration (or supervised) of the sucrose test: boil sugarcane juice with dilute HCl in the hot water bath, neutralise with sodium hydrogencarbonate until fizzing stops, and retest with Benedict's: brick-red. Learners record all results in a results table and compare with their predictions. They then compare Table 1 carbohydrate per 100 g: ugali 26 g, ripe banana 23 g, boiled sweet potato 20 g, sugarcane juice 18 g.	observations. Hold back the explanation: say "Hold that thought; we will explain it in the next phase." Have groups post their results on the class results table.		
Explain Phase  VIDEO: Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Carbohydrate Classification Practice Source: CK-12  Search ARES for similar readings	(20 min) Using the class results table, learners work in groups to build a claim-evidence-reasoning (CER) explanation for two questions: (1) 'What kind of carbohydrate is mainly in maize flour, and how do you know?' and (2) 'Why did sugarcane juice stay blue with Benedict's, but turn brick-red after boiling with dilute HCl?' The teacher then introduces carbohydrate structure with a bead model: one bead = monosaccharide (glucose, fructose; reducing sugars); two beads joined = disaccharide (sucrose, maltose); a long chain of many beads = polysaccharide (starch, glycogen, cellulose). Learners use the model to explain: starch in maize and sweet potato is a long chain that turns iodine blue-black; ripe banana has little starch left because starch has been converted to sugars, so iodine is faint and Benedict's is orange to brick-red; sucrose is a non-reducing disaccharide, so	Say: "Sugarcane juice is very sweet, yet Benedict's stayed blue. Talk in your group: how can that be?" WAIT TIME 10-15 seconds before taking answers. Cold-call 4 groups for their ideas and write them on the board without judging. Then introduce the bead model: "One bead is a monosaccharide. Two joined beads is a disaccharide. A long chain is a polysaccharide." Ask: "Where does sucrose fit? Where does starch fit?" Ask: "What did the acid do to the sucrose beads?" Use the sentence frame: 'My claim is... The evidence is... This shows that...'. Then ask: "Which food has the most carbohydrate per 100 g in Table 1, and what type of carbohydrate is it mostly?" Correct misconceptions: 'sweet always means reducing sugar' and 'all sugars react with Benedict's'. Link back: "Ugali is mostly starch. How could that help a runner like Kipchoge who needs about 2,200 kcal for a marathon?" Do not re-	Claim-Evidence-Reasoning (CER) with a bead model. Learners must connect their lab observations (evidence) to molecular structure (reasoning), which turns colour changes into an explanation of carbohydrate type. The physical model makes an invisible idea (monosaccharide to polysaccharide) concrete.	Look for CER answers that (1) cite specific colours (for example 'iodine blue-black means starch present'), (2) correctly classify maize flour as containing a polysaccharide (starch) and sugarcane juice as containing a non-reducing disaccharide (sucrose), (3) explain the HCl step as splitting sucrose into glucose and fructose. Use a quick exit check: each group holds up the number of beads for sucrose (2) and starch (many). Learners who write 'sugarcane has no sugar' need to be redirected to the difference between reducing and non-reducing sugar.

	Benedict's stays blue, but boiling with dilute HCl splits it into glucose and fructose, which are reducing sugars. Maltose is a reducing disaccharide. Learners use Table 1 to compare: ugali 26 g carbohydrate per 100 g, banana 23 g, sweet potato 20 g, sugarcane juice 18 g, and discuss why ugali, mostly starch, is a good store of energy for a runner: starch is stored in the plant as many glucose units, and in people as glycogen. Each learner writes a 2-3 sentence CER answer.	teach protein, lipids or enzymes; if asked how starch becomes sugar in the mouth say: "Great question; we will investigate that in Lesson 5."		
Driving Question Board (DQB) Creation  VIDEO: Hydrolysis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Carbohydrates Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(8 min) Learners return to the class DQB started in Lesson 1. Each group writes on sticky notes (a) one question answered by today's evidence and (b) one new question that today's evidence raised, for example: 'If ugali is starch, how does the body turn it into glucose?', 'Why does ripe banana taste sweeter than raw banana?', 'How is glycogen different from starch?' and 'Why did chewing ugali taste sweet?'. They place notes on the DQB, mark answered ones with a tick and cluster them under headings such as 'Carbohydrates', 'Other chemicals', 'Body processes'.	Say: "Look at the board. Which of our questions did today's tests help to answer? Which new questions do we have?" WAIT TIME 10-15 seconds while groups write. Cold-call 3 groups to read a new question aloud and say where it should go. Say: "Notice that several questions ask how the body breaks starch down. That will guide Lesson 5." Point to the main driving question: "How do simple foods like ugali, githeri and beans provide the chemicals athletes need?" and ask: "What part of this have we answered so far?"	Driving Question Board with question clustering. Learners see their own questions drive the unit, notice how evidence answers some questions and creates others, and take ownership of the investigation sequence.	Check that notes are (1) written as questions linked to today's evidence, (2) placed in a sensible cluster, and (3) include at least one question that goes beyond today's lesson (for example about digestion or glycogen). Count notes per group; groups that only repeat Lesson 1 questions are asked, 'What surprised you in today's tests?'
Model Building Phase  VIDEO: 03 Carbohydrates <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  HTML:	(12 min) Learners return to the single class model on chart paper (and their own initial model from Lesson 1). In groups they revise it by adding: (a) a carbohydrate section showing starch in ugali (maize) as a long chain of glucose units, with arrows to glucose as the energy source and glycogen as the store in muscle and liver; (b) the three carbohydrate groups with examples (monosaccharides: glucose, fructose; disaccharides:	Say: "Take out last lesson's model. What can you now add or change because of the evidence we collected today?" WAIT TIME 10-15 seconds before groups begin. Circulate and ask: "Which test result supports this part of your model?" Cold-call 3 groups: "Tell us one thing you added and the evidence for it." Ask: "What is still missing from our model if we want to explain Kipchoge's 2:01:09?" Record the gaps on the chart	Cumulative model revision with evidence labels. Requiring each addition to cite a specific test result links the model to evidence and shows learners how their explanation of the phenomenon grows lesson by lesson.	Look for models that (1) show starch as a chain of glucose units linked to energy, (2) correctly label the iodine and Benedict's colour changes, (3) distinguish reducing from non-reducing sugar, and (4) mark open questions. Use a 3-point check: Model includes evidence labels (yes/partly/no). Learners who draw starch as a single unit are prompted: 'What did the iodine test tell you about the size of starch?'

Carbohydrate Classification (Flexbook) Source: CK-12 Search ARES for similar readings	sucrose, maltose; polysaccharides: starch, glycogen, cellulose); (c) a small key for tests: iodine blue-black = starch; Benedict's blue to green to orange to brick-red = reducing sugar; sucrose needs boiling with dilute HCl first. They add Table 1 values (ugali 26 g, banana 23 g, sweet potato 20 g, sugarcane juice 18 g per 100 g). They also mark with a question mark what the model still cannot explain, such as how starch becomes glucose in the body, and the roles of protein and lipids. Each group shares one change they made and why, then completes the lesson summary table.	paper under 'Still to explain'. Close with the celebration: "You didn't just learn ABOUT science. You DID science!" and invite learners to note feeling accomplished, valuing the process, and feeling confident, with a thumbs-up check.		
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D. TEACHER REFLECTION

After the lesson, consider: Did learners record the actual colours they saw, or the colours they expected? Which groups were surprised that sweet sugarcane juice stayed blue, and did that surprise help them understand non-reducing sugar? Did any learners confuse 'reducing sugar' with 'all sugar' or think starch and sugar are unrelated? How well did the bead model help learners explain the HCl step? Was there enough time for the water bath, and were safety rules (goggles, no tasting) followed by every group? Which DQB questions about digestion should be highlighted for Lesson 5? What would you change in the grouping or timing next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that maize flour and sweet potato turned iodine blue-black, ripe banana gave only a faint brown-orange with iodine, and Benedict's gave blue for maize flour, green-yellow for sweet potato, orange to brick-red for ripe banana, and blue for sugarcane juice until it was boiled with dilute HCl and then gave brick-red.
What did I learn?	We learned that carbohydrates include monosaccharides (glucose, fructose), disaccharides (sucrose, maltose) and polysaccharides (starch, glycogen, cellulose); iodine turns blue-black with starch; Benedict's detects reducing sugars (glucose, fructose, maltose) but not the non-reducing sugar sucrose until it is split by boiling with dilute HCl. Table 1 shows 26 g carbohydrate per 100 g in ugali, 23 g in ripe banana, 20 g in boiled sweet potato and 18 g in sugarcane juice.
How does this explain the phenomenon?	We can explain that ugali is not plain: its maize starch is a long chain of glucose units that stores energy, and the tests give evidence for this. This helps explain how simple Kenyan foods supply the carbohydrate energy that athletes like Eliud Kipchoge need, but we still need to find out how the body breaks starch down and what proteins and lipids do.

LESSON 3 (40 minutes): Proteins: building and repairing the body

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the Biuret test to find which Kenyan foods contain protein and explain how proteins, as chains of amino acids, support growth and repair of muscle in athletes such as Eliud Kipchoge.
Knowledge	Proteins are long chains of amino acids. The Biuret test turns from blue to violet when protein is present. Beans, milk, egg white and omena give violet; ugali stays blue because it has too little protein to detect. Proteins supply amino acids for growth and repair of muscle. Table 1 protein per 100 g: omena 60 g, egg 13 g, beans 8 g, milk 3.3 g, ugali 2.5 g.
Skills	Planning and carrying out a food test safely, recording results in a table (building on Lesson 2), comparing test results with Table 1 data, constructing an explanation from evidence, and revising a model.
Attitudes	Curiosity about everyday foods, care and responsibility when handling reagents, respect for the ideas of group members, and appreciation of locally available protein foods such as beans, milk and omena.
Key Inquiry Question	What are the chemicals of life? How do chemicals of life contribute to the functioning of living organisms?
Purpose in Storyline	Lesson 1 named protein as one of the three chemicals of life. Lesson 2 taught carbohydrate tests and how to record results in a table. Lesson 3 adds the second chemical, protein, to the class model: it explains the repair and body-building part of the driving question. Lesson 4 will add lipids, and Lesson 5 will show that enzymes are proteins.
Safety Notes	Wear goggles during the Biuret test. Biuret reagent contains sodium hydroxide and copper sulfate, so wash any splash on skin immediately with plenty of water. Do not taste any lab samples. Use only a few drops of reagent. Wash hands after the practical. Dispose of the mixtures as the teacher directs. Keep omena and milk samples covered and wipe benches.

B. LESSON OVERVIEW

Learners start with Eliud Kipchoge, who hard training breaks down muscle in, and ask what the plain plate of ugali and beans gives him to rebuild it. They predict which foods contain protein, then run the Biuret test on beans, milk, egg white, omena and ugali, and record results in a table as in Lesson 2. They compare the colours with the Table 1 protein values, build a model of proteins as chains of amino acids, and explain growth and repair. The class adds new questions to the DQB and adds protein to the single class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Introduction to proteins and amino acids Source: Khan	(6 min) Learners recall Lesson 1, where protein was named as the chemical for body-building, and Lesson 2, where they tested foods and recorded results in a table. The teacher shows a picture or	Say: 'Last lesson we proved that ugali is full of starch. Today we ask: what repairs the athlete's muscles after he runs? Which foods do you think contain protein? Write your own ranking first,	Prediction and ranking with reasons (Asking Questions and Defining Problems). Writing a prediction before testing makes learners commit to ideas and expose their preconceptions. They	Look for: each learner has written a ranking with a reason (not just a list); groups have a prediction table ready; at least 4 groups can state a reason linked to the animal or body source of the food (e.g.

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Proteins Source: Khan Academy (English - US curriculum) Search ARES for similar readings	describes the plate of ugali, beans, sukuma wiki and milk eaten in Iten. Learners individually write: 'After a hard 2-hour run, Kipchoge's muscles are damaged. Which foods on the plate do you think contain protein to repair them? Rank beans, milk, egg, omena and ugali from most to least protein, and give a reason.' Then in groups of 4 they share and agree on one predicted ranking and sketch a prediction table with a column for 'Predicted' and a column for 'Biuret result'.	silently, for 2 minutes.' Walk around and note ideas without correcting. Say: 'Turn to your group and agree on one ranking.' After groups finish, cold-call 4 different groups: 'Which food did you put first, and what is your reason?' WAIT TIME 10 seconds after each question before calling on anyone. Record the predictions on the board and do not say which are right. Ask: 'Which food are you least sure about? Why?' Expect uncertainty about ugali and omena, and say: 'Good, today the evidence will help us decide.'	then judge these against the evidence, which makes the test results more meaningful.	'omena is fish, so it is meat-like'). Note learners who think ugali has the most protein because it is the main food, to address later.
Observe Phase VIDEO: Introduction to proteins and amino acids Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Proteins Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(14 min) Groups of 4 carry out the Biuret test on five prepared samples: mashed boiled beans in water, milk, egg white solution, crushed omena in water, and ugali mixed in water. Each learner takes a role: reagent handler, recorder, observer of colour, and time and safety monitor. Learners put about 2 cm of each sample in a test tube, add a few drops of Biuret reagent (or sodium hydroxide followed by a few drops of dilute copper sulfate), shake gently and watch for the colour. Blue stays blue (no protein detected); violet shows protein. They record observations in the results table and add a control tube of water. Expected results: beans, milk, egg white and omena turn violet; ugali stays blue; water stays blue. Learners then place Table 1 protein values beside their results (omena 60 g, egg 13 g, beans 8 g, milk 3.3 g, ugali 2.5 g per 100 g).	Before starting say: 'Goggles on, everyone. Biuret reagent contains sodium hydroxide, so no tasting, and wash any splash straight away.' Check goggles at every group. Model the test with beans for 1 minute. Say: 'Why do we test water as well?' and WAIT 10 seconds, then cold-call 2 learners to say it is a control showing the reagent alone stays blue. Circulate and ask: 'What colour do you see, and what does it tell you about this food?' Pause 10 seconds. Remind groups: 'Record exactly what you see, blue or violet, before you look at the Table 1 numbers.' Once results are in, give each group the Table 1 protein values and say: 'Now compare. Which food has the most protein? Which has the least? Does the colour change match the numbers?' Ask the group with ugali: 'Ugali has 2.5 g per 100 g, but stays blue. Why might that be?' (Too little to detect at the classroom concentration.) Cold-call 3 groups to report one result each.	Collecting and organising data in a table (Planning and Carrying Out Investigations; Analysing and Interpreting Data). Recording first and comparing with Table 1 second keeps learners from guessing what they 'should' see, and linking colour to numbers shows that a test result is evidence about the amount of a chemical.	Observable indicators: goggles worn by all learners; a control tube included; the results table has all five foods plus water with correct colours (beans, milk, egg white, omena violet; ugali blue); learners state that violet means protein present; at least 3 groups notice that omena (60 g) is the richest protein food in Table 1. Note learners who say ugali has no protein at all instead of 'too little to detect'.
Explain Phase	(10 min) Learners use their results	Say: 'Protein is a very long chain	Claim-Evidence-Reasoning with a	Look for: explanations that state

 VIDEO: Introduction to amino acids <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Amino Acids (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	and Table 1 to answer: 'What does the evidence tell us about where protein comes from in the Iten diet?' The teacher introduces the idea that protein is a long chain of amino acids. Learners model this using paper clips or beads of different colours linked in a chain (amino acids joined into a protein), then 'break' the chain to show amino acids released during digestion and 'rebuild' a different chain to show new muscle protein. In groups they write a claim-evidence-reasoning explanation: 'Beans, milk and omena help an athlete's muscles recover because...' using the Biuret colours and Table 1 values, and they explain how protein supports growth (new cells in young athletes) and repair (damaged muscle fibres after running). They compare with Lesson 2: 'ugali is mainly starch for energy, but beans add protein.' They then explain why githeri (maize and beans, 6 g protein per 100 g) is a good combination.	made of small units called amino acids. Your paper-clip chain is one protein. When Kipchoge eats beans, his body breaks the chain into single amino acids, and then joins them again into new muscle protein.' Ask: 'What does the body do with the amino acids?' WAIT 10 seconds, cold-call 3 learners. Prompt with: 'After a long run muscle fibres are damaged. What do they need to be repaired?' Then say: 'Write a claim, then evidence from our Biuret colours and Table 1, then your reasoning.' Provide the frame: 'I claim... because the Biuret test showed... and Table 1 shows... which means...'. Circulate and challenge: 'Does Table 1 say ugali is a good protein food? So where does the athlete get most protein from this plate?' Cold-call 3 groups to read their claim aloud, and ask another group to say whether the evidence supports it. Correct misconceptions only through questions, for example 'Is protein the main energy store, or is that carbohydrate? What did we learn in Lesson 2?'	physical model (Constructing Explanations; Developing and Using Models). Building the chain makes an invisible molecule concrete, and the CER frame forces learners to tie the explanation to their own test results rather than to memory.	protein is a chain of amino acids; learners who link amino acids to repair of muscle and growth, citing beans, milk, egg or omena; use of at least one Biuret result and one Table 1 number in the evidence; correct statement that ugali is mainly a starch food with little protein. Mark as secure if learners can say why githeri supplies more protein than ugali (6 g vs 2.5 g per 100 g). Watch for 'protein gives energy first' misconceptions.
Driving Question Board (DQB) Creation  VIDEO: Tertiary structure of proteins <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Proteins	(5 min) Learners return to the class DQB started in Lesson 1. Each group writes on sticky notes (a) one question from the protein evidence that is now answered and (b) one new question that has appeared, such as 'Why do athletes in Iten eat omena or beans if ugali already fills them?', 'Do we need protein every day?', 'How does the body break the protein chain?', or 'What are enzymes made of?'. They stick the notes on the board in the sections 'Energy', 'Growth and repair', or 'Still wondering'. The class chooses one new question to	Say: 'Go back to our driving question about how simple foods give the chemicals athletes need. What can we now answer about protein? What new question did the test make you ask?' Give each group 2 minutes. WAIT 10 seconds before collecting. Cold-call 4 groups to read a note and place it. Ask: 'Which of these questions will help us explain how the body gets energy and repairs itself?' Point out that the enzyme question will be taken up in Lesson 5 and the lipid question in Lesson 4, without teaching them now. Mark answered questions with a	Driving Question Board (Asking Questions). Linking new questions to evidence shows learners that science is a cycle of answering and asking, and gives them ownership of the unit direction.	Check that each group has at least one answered and one new question; that new questions connect to the phenomenon rather than being unrelated; and that at least 2 questions are about how or why (for example how proteins are broken down). Note which groups still confuse protein with energy storage.

Source: Khan Academy (English - US curriculum) Search ARES for similar readings	follow in a later lesson.	tick and a short note, for example 'Beans, milk, egg, omena = protein: Biuret violet'.		
Model Building Phase  VIDEO: Peptide bond formation <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Proteins Source: Khan Academy (English - US curriculum) Search ARES for similar readings	(5 min) Learners add protein to the single class model on chart paper, which already shows carbohydrates (starch in maize in ugali) as the energy chemical from Lessons 1 and 2. A representative from each group adds: foods rich in protein (beans, milk, egg, omena), a drawing of a chain of amino acids, an arrow from protein food to 'amino acids' to 'muscle growth and repair', and a note about the Biuret test (blue to violet). They label how this part explains the athlete's recovery after running, and mark a question mark where lipids and enzymes will be added. Learners then complete the summary table prompt individually in their notebooks, and the teacher closes the lesson.	Say: 'Our model must now explain more of the plate. What does protein do that starch does not? Add it where it belongs.' Hand a marker to 4 different learners to add the four parts (foods, chain, arrow to muscle, test colour). WAIT 10 seconds before each, and ask the class: 'Does this addition match our evidence?' Ask: 'Which parts of the plate of ugali, beans, sukuma wiki and milk now have a job in the model?' Cold-call 3 learners. Finish with the template closing: 'You didn't just learn ABOUT science—you DID science!' and tick the celebration checklist: students feel accomplished, value the process and are confident.	Cumulative model revision (Developing and Using Models). Adding protein to the same model, rather than starting a new one, shows learners how each investigation builds the whole explanation of the phenomenon.	Check that the model includes protein foods, amino acids, and growth and repair as linked parts, and that carbohydrate is still shown as the energy source. Look for learners explaining each addition using evidence (Biuret colours, Table 1). Individual summary tables should show correct results: beans, milk, egg white and omena violet; ugali blue.

D. TEACHER REFLECTION

Which learners could explain, using Biuret colours and Table 1 values, why ugali stays blue yet beans turn violet? Did groups confuse protein with the main energy store? How well did the paper-clip chain help learners see amino acids joined together? Were safety routines with the reagent followed by every group? Which new DQB questions (for example about enzymes and lipids) should be highlighted for Lessons 4 and 5? What would I change to give quieter learners more speaking time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We tested beans, milk, egg white, omena and ugali with Biuret reagent. Beans, milk, egg white and omena turned violet; ugali stayed blue. Table 1 shows omena has the most protein (60 g per 100 g), then egg (13 g), beans (8 g), milk (3.3 g) and ugali (2.5 g).
What did I learn?	Proteins are long chains of amino acids. Biuret turns blue to violet when protein is present. Foods like beans, milk, egg and omena are good protein sources, while ugali has too little to detect.
How does this explain the	Athletes in Iten eat beans, milk and omena with ugali because the proteins are digested to amino acids, which the body uses to

phenomenon?	grow and repair muscle damaged by running, while ugali supplies starch for energy. This adds protein to our model of how a plain plate supports Kipchoge's performance.
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LESSON 4 (80 minutes (double lesson)): Lipids: stored energy and insulation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence that lipids are a third chemical of life in Kenyan foods. They use the brown paper and emulsion tests, and use Table 1 fat values to show why lipids store more energy per gram than carbohydrates, and so explain part of how simple foods fuel athletes such as Eliud Kipchoge.
Knowledge	Lipids (fats, oils and waxes) are made of fatty acids and glycerol. Fats are solid and oils are liquid at room temperature. Lipids are insoluble in water but dissolve in ethanol. The brown paper test gives a translucent spot that stays after drying. The emulsion test gives a cloudy white layer. Lipids give 9 kcal per g, compared with 4 kcal per g for carbohydrate (Lesson 2) and for protein (Lesson 3). Lipids store energy and insulate the body.
Skills	Planning and carrying out an investigation (NGSS SEP 3): performing the brown paper and emulsion tests safely. Analysing and interpreting data (SEP 4): comparing Table 1 fat values and energy. Using mathematics (SEP 5): applying $\text{Energy} = 4 \times C + 4 \times P + 9 \times F$ to groundnuts, mandazi and nyama choma. Constructing explanations (SEP 6) and developing models (SEP 2): adding lipids to the class model.
Attitudes	Curiosity about how everyday Kenyan foods such as groundnuts and mandazi contain hidden chemicals. Respect for peers' ideas during group discussion. Responsibility in handling ethanol and food samples. Appreciation that science explains familiar foods and diet choices.
Key Inquiry Question	1. What are the chemicals of life? 2. How do chemicals of life contribute to the functioning of living organisms?
Purpose in Storyline	Lesson 1 named lipids as one of three chemicals of life. Lessons 2 and 3 taught the testing routine (observe, record in a table, conclude) for carbohydrates and proteins. Lesson 4 adds lipids with two new tests and the first evidence that fat holds a lot of energy per gram. This prepares Lesson 5 (enzymes that break food down) and Lesson 6 (comparing Meals A, B and C using fat values and energy).
Safety Notes	Ethanol is flammable: keep it away from flames and use only small amounts, with no heating of ethanol in the room. Wear goggles during the emulsion test. Do not taste any lab sample, including groundnuts and mandazi used for tests. Check for groundnut allergy before the lesson, and let affected learners record data and handle only the cooking oil and mandazi samples, or observe. Wash hands after testing. Wipe up oil spills at once to prevent slipping. Dispose of waste in the bin and not down the sink.

B. LESSON OVERVIEW

Learners start by predicting which Kenyan foods contain lipids and why a small handful of groundnuts feels so filling. They then test groundnuts, cooking oil, mandazi, nyama choma and maize flour with the brown paper and emulsion tests, recording results in a table as in Lessons 2 and 3. In the Explain Phase they use Table 1 and the Atwater factor of 9 kcal per g to compare the energy in 100 g of groundnuts, mandazi and nyama choma with ugali, linking lipids to energy storage and insulation. The class adds a new question to the Driving Question Board (DQB) and adds lipids to the single class model on chart paper, so that the plain plate of ugali and beans is seen as one that holds several chemicals of life.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Lipid overview Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Lipids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the plain plate of ugali and beans from Lesson 1 and the question of how it fuels Eliud Kipchoge, who trains near Iten. The teacher shows five samples on a tray (roasted groundnuts, cooking oil, mandazi, nyama choma, dry maize flour). Individually, learners write on a sticky note or in their books: (a) which samples they think contain lipids, and (b) what evidence they could see or feel to tell. They then discuss in groups of 4 and agree a ranking from 'most lipid' to 'least lipid'. Learners record their predictions in a prediction table for later checking.	Say: "In Lesson 1 we named lipids as one of the three chemicals of life. Today we find out how we can prove a food has them." Hold up the groundnuts and say: "A small handful of these keeps a runner going for hours. Write your own prediction first." Give 2 minutes of silent writing. Then ask: "Which of these five foods do you think has the most lipid? What makes you say so?" WAIT TIME 10-15 seconds before taking any answer. Cold-call 4 learners from different groups for their rankings and reasons. Ask 2 learners: "Where have you seen oil leave a mark on paper or a plate at home, for example after eating mandazi?" Record all predictions on the board without judging. Say: "Keep your predictions. We will check them with evidence."	Predict-Observe-Explain (prediction stage) with Think-Pair-Share. Writing a prediction first lets learners commit to their own ideas, and everyday experience (oil marks from mandazi) becomes the starting point. Comparing rankings in groups surfaces differences that create a need for a test, so the later evidence feels meaningful.	Check that at least 80% of learners wrote a ranking with a reason in their books. Look for: Do learners name oil marks, greasy feel or taste as evidence (acceptable everyday ideas)? Do some think maize flour has lipids, or that only oils count and nyama choma does not? Note these ideas on a tally to revisit in the Observe Phase. Listen for whether they connect to the earlier food tests (colour changes) in Lessons 2 and 3.
Observe Phase  VIDEO: Molecular structure of triglycerides (fats) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Lipids (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 4, learners carry out two tests on each sample: groundnuts, cooking oil, mandazi, nyama choma and maize flour (comparison sample). BROWN PAPER TEST: rub a small amount of each sample on a labelled piece of brown paper, with water as a control. Leave to dry for 10 minutes, then hold each up to the light and note whether a translucent (see-through) spot remains. EMULSION TEST: crush a small piece of groundnut (or put 5 drops of cooking oil) in a dry test tube with about 2 cm³ of ethanol, shake well so the lipid dissolves, then pour the clear liquid into a second tube holding 2 cm³ of water. A cloudy white layer shows lipid. Learners record results in a table with columns: food, brown paper spot (translucent / no spot), emulsion test (cloudy white /	Before starting, hold up the safety reminders: "Goggles on. Ethanol stays away from flames. No tasting any sample, not even the groundnuts." Demonstrate once how to rub a sample on the paper and how to pour the ethanol layer into the water. Circulate with a checklist and ask: "What do you see when you hold the paper up to the light? Describe it exactly." WAIT TIME 10-15 seconds after each question before prompting. Ask: "Why do we test water on the paper too?" Ask one group: "Maize flour is the main food in ugali. What does your result tell you about the lipid in it?" Cold-call 3 groups to report their table row for nyama choma and mandazi and ask: "Did that match your prediction?" Challenge the idea that only liquid oils contain lipid: "Nyama choma is solid meat. Why	Evidence table with controlled comparison. Using a water control and a maize flour comparison helps learners see that the test result comes from the lipid and not from the paper getting wet. Reusing the same results-table routine from Lessons 2 and 3 builds transfer of method. Comparing two different tests on the same samples shows that more than one line of evidence supports the same claim.	Observable indicators: learners wear goggles and keep ethanol away from flames (tick on checklist); the results table has all four columns filled for five foods; learners correctly state that the spot staying after drying means lipid is present; learners can say that maize flour gave no translucent spot. Common error to watch: reading a wet spot as a positive result before it has dried; reminding them to wait the full 10 minutes. Ask 3 learners to point to the control spot and explain what it shows.

	clear), conclusion. Expected results: groundnuts, cooking oil, mandazi and nyama choma leave a translucent spot that stays after drying (lipid present). The emulsion test gives a cloudy white layer for groundnuts and cooking oil. Maize flour gives no translucent spot. Learners compare these with their predictions and with the Table 1 fat values (groundnuts 49 g, mandazi 15 g, nyama choma 15 g, ugali 0.6 g per 100 g). Learners must also explain why the water control spot disappears but the oil spot stays.	does it leave a spot too?" Do not give the explanation yet; say "Keep that question for the next phase." Collect the class results on a board table.		
Explain Phase  VIDEO: Lipid overview Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Lipids Practice Source: CK-12  Search ARES for similar readings	Learners use their test results and Table 1 to build an explanation. Step 1: In groups, they list the foods that tested positive and look at the fat per 100 g in Table 1: groundnuts 49 g, mandazi 15 g, nyama choma 15 g, ugali 0.6 g. Step 2: Learners are told that lipids (fats, oils and waxes) are made of fatty acids and glycerol, that fats are solid and oils are liquid at room temperature, and that waxes are another type of lipid, for example on plant leaves and in beeswax. Step 3: Learners calculate the energy in 100 g of groundnuts, mandazi and nyama choma using $\text{Energy} = 4 \times C + 4 \times P + 9 \times F$, and check their answers: groundnuts $64 + 104 + 441 = 609$ kcal; mandazi $180 + 24 + 135 = 339$ kcal; nyama choma $108 + 135 = 243$ kcal. Step 4: They compare how much of the energy comes from fat in groundnuts (441 of 609 kcal), and compare with ugali at 119.4 kcal per 100 g (from Lesson 2: mainly starch, 104 kcal from carbohydrate). They state that 1 g of lipid gives 9 kcal while 1 g of carbohydrate gives 4 kcal, so fat	Say: "We have evidence that lipids are in four of our foods. Now we explain what lipids are and why they matter for our runner." Introduce structure briefly using the chart: "A lipid is built from fatty acids and glycerol, like a small frame holding three tails." Ask: "Fats are solid and oils are liquid at room temperature. Name one of each from your kitchen at home." Cold-call 3 learners. Model the first calculation on the board (groundnuts), then let groups do mandazi and nyama choma. Circulate and ask: "Which part of the sum gives the fat energy? How did you get 441?" WAIT TIME 10-15 seconds. Ask: "Compare 100 g of groundnuts at 609 kcal with 100 g of ugali at 119.4 kcal. What do you notice?" Cold-call 4 groups. Ask: "Eliud Kipchoge's marathon costs about 2,200 kcal and his stored glycogen is about 2,000 kcal. Where would extra energy storage come from, and why might too much fat be a problem for running at 52 kg?" Accept reasoned answers and push for evidence: "Which number supports	Claim-Evidence-Reasoning (CER) with quantitative comparison. Calculating energy from fat values turns the idea of 'stored energy' into a number that learners can compare (9 kcal per g against 4 kcal per g), so the explanation is based on data. Linking each calculation back to the 2,200 kcal marathon and 2,000 kcal glycogen keeps the phenomenon central. Naming the structure (fatty acids and glycerol) after the observations ensures the vocabulary explains evidence the learners already have.	Check the calculations: groundnuts 609 kcal, mandazi 339 kcal, nyama choma 243 kcal (accept if shown with working). Success indicator: learners write that lipid gives 9 kcal per g and that this is more than twice the 4 kcal per g of carbohydrate. Check that the CER paragraph names a result (translucent spot or cloudy white layer), cites a Table 1 number, and states a role (energy storage or insulation). Exit prompt on a slip: 'Name the two parts of a lipid and one role.' Expect 'fatty acids and glycerol' and 'energy store' or 'insulation'. Re-teach groups that confuse lipids with carbohydrates or who use 4 instead of 9 for fat.

	holds more than twice the energy per gram ($9 / 4 = 2.25$). Step 5: Learners explain roles: lipids store energy in the body in fat tissue, and lipids insulate, with a layer of fat under the skin reducing heat loss, and they link this to why Kenyan athletes need some fat but their main fuel is carbohydrate, with glycogen about 2,000 kcal against the 2,200 kcal marathon cost. Learners then write a claim-evidence-reasoning paragraph: 'Groundnuts contain lipid because...' and one sentence on why nyama choma and mandazi are also lipid foods.	that?" For insulation, ask: "Why might Iten's cool mornings matter for body fat?" Do not introduce enzymes or digestion of fats; say "We will see in Lesson 5 how enzymes break food down." Give sentence frames for the CER paragraph: "The evidence is... This shows..."		
Driving Question Board (DQB) Creation  VIDEO: Molecular structure of triglycerides (fats) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Lipids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the class DQB. In groups, they write new questions on sticky notes arising from today's evidence, for example: 'If groundnuts have so much energy, why do athletes eat mostly ugali?' or 'How does the body use stored fat when running?' They place each note on the board, then place one sticky note with an answer they can now give from Lesson 4 on an existing question about energy. They sort questions into: answered now, partly answered, and still to investigate (to be taken up in Lessons 5 and 6).	Say: "Look at the board. Which of our questions did lipids help us answer today?" WAIT TIME 10-15 seconds. Cold-call 3 groups to move or mark one note. Ask: "What new question has today's evidence made you ask?" Collect 4 new questions and read them aloud. Highlight those about digestion and about which meal is better for a runner, and say: "Lesson 5 helps with how food is broken down, and Lesson 6 will use data to compare meals." Check that the notes are specific and testable rather than 'What is a lipid?'. Prompt: "What would you need to measure or observe to answer your question?"	Driving Question Board as a record of growing understanding. Having learners link evidence to existing questions and write new ones keeps ownership of the learning with them and shows how each lesson moves the inquiry on towards the driving question about how simple foods provide energy, growth and repair.	Count how many groups add at least one new question and at least one answer link. Look for questions that mention a number or a food (for example 'Why does groundnut have 49 g fat?'). Note learners who connect lipids to the athlete (energy, insulation) and those who only ask definition questions. Use this to plan prompts for Lesson 5.
Model Building Phase  VIDEO: Cellular respiration <i>Related topic: not an exact match for this lesson.</i> Source: Khan	Learners revisit the single class model on chart paper (started in Lesson 1, with carbohydrates added in Lesson 2 and proteins in Lesson 3). Each group sends a representative to add lipids: a box for 'lipids (fats, oils, waxes)' showing groundnuts, cooking oil, milk fat and nyama choma as	Say: "Our model must show what we have evidence for. What new part do we add today?" Ask each group to propose one addition and justify it with data: "Which test or number shows that?" Cold-call 3 groups. Use colour coding: carbohydrate, protein and lipid each in their own colour.	Cumulative modelling. Adding lipids to the same class model shows how the three chemicals of life work together in one plate of food. Citing evidence for every addition keeps the model tied to data, and noting what is missing sets up the next lesson on enzymes.	Check that each learner's model includes lipids with sources, fatty acids and glycerol, 9 kcal per g, and two roles (energy store and insulation). Look for evidence labels. Check the written sentence for a correct link to the athlete. Identify learners who still place lipids as the main fuel without

Academy (English - US curriculum) Search ARES for similar videos HTML: Lipids (Flexbook) Source: CK-12 Search ARES for similar readings	sources, 'fatty acids and glycerol' as building parts, an arrow labelled '9 kcal per g' to 'energy store', and a second arrow to 'insulation under the skin'. They also add a line showing how lipids connect to the other two chemicals in the plate of ugali, beans, sukuma wiki and milk, and write a brief note on the evidence (brown paper and emulsion tests, Table 1). Learners then update their own copy of the model in their books and write one sentence: 'Lipids help a runner like Kipchoge by...'. Close with the celebration: learners reflect on what they did and feel confident about.	Challenge: "Does our model explain why ugali alone can fuel a marathon? What is still missing?" WAIT TIME 10-15 seconds. Record gaps such as 'how the body breaks food down' for Lesson 5. Finish with the tick-list: "Do you feel accomplished? Did you value the process? Do you feel confident?" Close with: "You didn't just learn ABOUT science—you DID science!"		reference to carbohydrate and glycogen, and discuss with them using the numbers 2,200 kcal and 2,000 kcal.
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D. TEACHER REFLECTION

Which learners used numbers (translucent spot, 609 kcal, 9 kcal per g) rather than everyday ideas to justify claims? Did any hold the misconception that fat is always bad, or that only liquid oils are lipids, and how did the nyama choma and mandazi results address this? Was the 10-minute drying time managed well? Were safety practices with ethanol followed and groundnut allergies handled? Did learners link lipids back to Kipchoge's energy needs? Which DQB questions should open Lesson 5 on enzymes?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Groundnuts, cooking oil, mandazi and nyama choma left a translucent spot on brown paper that stayed after drying, and groundnuts and cooking oil gave a cloudy white layer in the emulsion test. Maize flour left no translucent spot. Table 1 shows groundnuts have 49 g fat per 100 g, mandazi 15 g, nyama choma 15 g and ugali only 0.6 g.
What did I learn?	Lipids (fats, oils and waxes) are made of fatty acids and glycerol. Lipids give 9 kcal per g, more than twice the 4 kcal per g of carbohydrate. They store energy and insulate the body.
How does this explain the phenomenon?	Groundnuts have 609 kcal per 100 g (64 + 104 + 441) compared with 119.4 kcal for ugali, because fat holds 9 kcal per g. Lipids in foods like milk, groundnuts and cooking oil add stored energy and insulation to a runner's body, but the main fuel for the marathon (about 2,200 kcal) is carbohydrate stored as glycogen (about 2,000 kcal).

LESSON 5 (80 minutes (double lesson)): Enzymes: Why Chewed Ugali Tastes Sweet

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how temperature and pH affect the rate at which saliva amylase breaks down starch, and use the data to explain how enzymes help the body turn simple foods such as ugali into energy for athletes like Eliud Kipchoge.
Knowledge	Enzymes are protein catalysts (made of amino acids, from Lesson 3) that speed up the breakdown of big food molecules into small ones. Amylase breaks starch (Lesson 2) into maltose, a reducing sugar. Enzymes have an optimum temperature (about 37 C) and pH (near neutral, 6.8). Extreme heat or pH denatures an enzyme permanently. Each enzyme is specific to one substrate.
Skills	Planning a fair test and controlling variables; using the iodine test and Benedict's test; measuring time; calculating relative rate = 60 / time in seconds; drawing and interpreting line graphs; using data as evidence for an explanation (NGSS practices: planning and carrying out investigations, analysing and interpreting data, constructing explanations).
Attitudes	Curiosity about everyday food experiences; honesty in recording results; teamwork and respect for others' ideas in groups of 4; appreciation that traditional Kenyan foods work with the chemistry of the body.
Key Inquiry Question	KICD: How do chemicals of life contribute to the functioning of living organisms? Lesson question: Why does chewed ugali taste sweet, and what conditions does the enzyme need to work?
Purpose in Storyline	Lessons 2 to 4 showed that ugali and beans contain carbohydrates, proteins and lipids. Lesson 5 answers the next question: how can the body use big molecules such as starch? Enzymes break them into small molecules that can be absorbed. This adds digestion to the class model and prepares Lesson 6, where learners compare meals using evidence.
Safety Notes	Goggles on for Benedict's test and hot water baths. Use only your own saliva or teacher-supplied amylase solution, and do not share. Never taste any lab sample; the ugali tasting is done only with clean, teacher-approved ugali before the lab begins, outside the lab area. Handle hot water with care and keep it away from the edge of the bench. Dilute HCl and NaOH (if used for pH) are prepared by the teacher only, and spills are wiped up straight away. Wash hands after the practical.

B. LESSON OVERVIEW

Learners begin by predicting what happens when ugali is chewed for a long time. They then run a fair test on amylase: 1% starch solution plus saliva amylase at 0, 20, 37, 60 and 80 C and at pH 2, 4, 6.8, 9 and 12 (at 37 C). Time for iodine to stop turning blue-black is recorded, and relative rate = 60 / time is calculated. Data show 37 C is 5 times faster than 20 C (1.0 / 0.2), and no starch is cleared at 0, 60, 80 C or at pH 2 and 12 within 600 s. Learners explain denaturation and specificity, update the DQB and add enzymes to the class model on chart paper. The phenomenon returns to Iten, where an athlete's body temperature of about 37 C keeps amylase working on the ugali he eats.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are given a small piece of clean, teacher-approved ugali	Say: 'Ugali is mostly starch, and you learned in Lesson 2 that	Predict-Observe-Explain (POE) with a personal experience hook:	Look for: a written prediction that links sweetness to starch changing

 VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Enzymes in the Digestive System (Flexbook) Source: CK-12  Search ARES for similar readings	(or shown a photograph if there is no food) and asked to chew it slowly for one minute without swallowing. They record in notebooks what happens to the taste. Then they individually predict: (1) Why does it taste sweeter when chewed for longer? (2) What would happen to this change if the ugali were very cold or very hot? They draw a quick sketch of a prediction graph of reaction rate against temperature, linking to how an athlete such as Kipchoge in Iten digests his ugali.	starch does not give a sweet taste. So where does the sweetness come from?' Give WAIT TIME of 10-15 seconds in silence after the question. Cold-call 4 learners to share the taste observation, and ask 'What evidence from your mouth tells you that something changed?' Do not confirm anything yet. Ask 3 learners to share their prediction graph and write the different shapes on the board. Say: 'Keep your prediction. By the end of the lesson we will check it against data.'	learners commit to a prediction using a familiar experience (the taste of chewed ugali). Recording a prediction first exposes prior ideas and creates a reason to check the evidence later.	(even if it lacks the word 'enzyme'); a sketched graph with some shape (rising, peak or flat). Tally on the board how many learners predict that more heat gives faster change. Misconception to note: 'Hotter always means faster', or 'saliva only wets the food'.
Observe Phase  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzymes in the Digestive System Practice Source: CK-12  Search ARES for similar readings	In groups of 4, learners carry out the amylase investigation. Roles: timekeeper, pipette handler, recorder, safety officer. Step 1: Label test tubes with temperature (0, 20, 37, 60, 80 C). Add 5 cm³ of 1% starch solution and place each in its water bath (ice, room temperature, 37 C, 60 C, 80 C) for 3 minutes. Step 2: Place drops of iodine on a white tile. Step 3: Add 1 cm³ of saliva amylase to a tube, mix, and every 30 seconds transfer a drop to the iodine. Record the time when the iodine no longer turns blue-black, or stop at 600 s. Step 4: For pH, repeat at 37 C with buffers at pH 2, 4, 6.8, 9 and 12. The class pools the results in a table: temperature 0 C not cleared in 600 s (rate 0); 20 C 300 s (0.2); 37 C 60 s (1.0); 60 C not cleared (0); 80 C not cleared (0). pH 2 not cleared (0); pH 4 240 s (0.25); pH 6.8 60 s (1.0); pH 9 180 s (0.33); pH 12 not cleared (0). Groups also test a sample at the end of the 37 C tube with Benedict's (hot water bath) and see an orange to brick-red colour, showing a reducing sugar (maltose) was made.	Before starting, say: 'What must we keep the same so this is a fair test?' Cold-call 3 learners to name the control variables (volume of starch, volume of amylase, concentration, same drop size). Remind everyone: 'Goggles on. No tasting. Hot water baths are only for the safety officer to move.' Circulate and ask each group: 'What colour do you expect at the end if starch is gone?' Give WAIT TIME of 10-15 seconds before taking answers. Stop groups that cannot clear the starch after 600 s and say: 'No change after 600 s is still a result. Record it as not cleared.' Give the class table at the end so all groups have the same data and any outliers are discussed.	Collaborative investigation with a class data pool: pooling results lets learners compare many conditions and see patterns that a single group could not. Role cards keep every learner involved in the evidence collection.	Observe: Are variables controlled (same volumes, same temperature before mixing)? Does the recorder write times in seconds and write 'not cleared' correctly? Does the group's Benedict's result match the class result? Check that learners use the iodine colour correctly (blue-black = starch present). Use a quick checklist for each group, noting safety behaviour.

Explain Phase  VIDEO: Introduction to kinetics <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Enzymes in the Digestive System (Flexbook) Source: CK-12  Search ARES for similar readings	Learners calculate relative rate = $60 / \text{time in seconds for each condition}$ (for example $60 / 300 = 0.2$ at 20 C and $60 / 60 = 1.0$ at 37 C) and plot two line graphs (rate against temperature, rate against pH). They compare with their predictions. They calculate the comparison $1.0 / 0.2 = 5$, so the reaction at 37 C is 5 times faster than at 20 C. In groups, they use a Claim-Evidence-Reasoning (CER) frame to answer: 'Why does chewed ugali taste sweet, and why does the enzyme stop working at 60 C and pH 12?' Teacher introduces the key ideas: an enzyme is a protein (made of amino acids, Lesson 3) that acts as a catalyst; amylase breaks starch into maltose (a reducing sugar, Lesson 2); the optimum is about 37 C and near pH 6.8; heat or extreme pH changes the shape of the enzyme permanently, so it is denatured; each enzyme is specific, so amylase works on starch, not on protein or fat. Learners link this to a human body at about 37 C in lten, where the ugali is broken down so the glucose can fuel running.	Ask: 'At which temperature was the rate fastest? How do you know?' WAIT TIME of 10-15 seconds, then cold-call 4 learners. Ask: 'Between 37 C and 60 C the rate drops from 1.0 to 0. Is the enzyme just resting, or has something changed permanently?' Cold-call 3 groups. Use a model: a key fitting a lock (starch fits amylase, protein does not) to show specificity, and crumple a paper key to show denaturation. Ask: 'If we cool the 60 C tube back to 37 C, would the enzyme work again? Why not?' Give the vocabulary: optimum, denatured, specific. Then say: 'Using only the table, what is your claim?' and check that each CER statement includes at least two numbers.	Claim-Evidence-Reasoning with data graphing: learners turn the numbers into a graph, find the peak, and justify claims with specific values. The lock-and-key analogy helps learners visualise the shape of the active site so they can reason about specificity and denaturation.	Success indicators: graphs have labelled axes with units and a smooth curve peaking at 37 C and pH 6.8; the rate calculation is correct (0.2, 1.0, 0.25, 0.33); the CER cites at least two data values (e.g. 'rate 1.0 at 37 C compared with 0.2 at 20 C, 5 times faster') and uses the word denatured correctly. Misconceptions to catch: 'cold kills the enzyme' (at 0 C the enzyme is slowed, not denatured), and 'enzymes are living'.
Driving Question Board (DQB) Creation  VIDEO: Introduction to kinetics - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos	Learners revisit the class DQB. Each group writes on sticky notes (a) one question that is now answered by Lesson 5, (b) one new question. Examples: 'Why does chewed ugali taste sweet?' (moved to answered: amylase breaks starch to maltose at about 37 C), 'Do we have enzymes for proteins and fats too?', 'Does Kipchoge's fever or very hot weather affect digestion?', 'Can a person with a very acidic stomach digest starch?'. They place the notes on the DQB and link them to	Say: 'Which of our earlier questions can we now answer with evidence from today?' WAIT TIME of 10-15 seconds. Cold-call 3 groups to move a note and justify using a number from Table 3. Ask: 'What new question has this lesson created that we will need for Lesson 6?' Collect at least 4 new questions and group them into 'enzymes', 'food and energy' and 'meals'. Praise questions that are testable.	Driving Question Board as a public record of growing understanding: moving answered questions and adding new ones makes the learning trail visible and ties each lesson back to the phenomenon.	Check that sticky notes are specific and testable rather than 'What is an enzyme?'. At least one question per group should connect enzymes to the athlete's diet. Note which learners cite evidence when moving a question to 'answered'.

 EXERCISE: Enzymes in the Digestive System Practice Source: CK-12  Search ARES for similar readings	the main driving question about how simple foods supply energy, growth and repair.			
Model Building Phase  VIDEO: Competitive inhibition <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzyme structure and catalysis <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners add digestion to the single class model on chart paper. Using arrows and labels, they show: ugali (starch) in the mouth, saliva amylase (a protein enzyme) acting at about 37 C and near pH 6.8, starch broken into maltose, then into glucose that enters the blood for energy and is stored as glycogen (about 2,000 kcal store for a marathon costing about 2,200 kcal). They add a note box: 'Heat above 60 C or pH 2 or 12 denatures enzymes' and 'Amylase is specific to starch'. They also connect beans and milk (proteins) and groundnuts (lipids) as other foods that need their own enzymes. Each group states one revision to the initial model from Lesson 1 and writes it on a sticky note.	Say: 'Our model from Lesson 1 showed food going in and energy coming out. What is missing in between?' WAIT TIME of 10-15 seconds. Cold-call 4 learners to add one part with a reason from today's data. Ask: 'Where in our model would Kipchoge's body be at 37 C, and why does that matter?' Guide the class to draw the digestion step between food and energy without re-teaching carbohydrates, proteins or lipids. Close with the celebration: ask learners to tick the feelings 'I feel accomplished', 'I value the process', 'I am confident'. Say: 'You didn't just learn ABOUT science. You DID science!' Preview Lesson 6: 'Next we will use all our evidence to decide which meal best supports the athlete, citing numbers.'	Cumulative model revision: adding a new component (enzymes and digestion) to a single class model shows how each piece of evidence changes the explanation of the phenomenon. Learners justify each addition with data.	Look for: model includes amylase, starch to maltose to glucose, an optimum label with temperature 37 C and pH 6.8, and a denaturation note. Each addition is justified by a data point. Exit check: ask each learner to write one sentence beginning 'Amylase is affected by temperature because...'. Responses should mention shape changing or denaturing.

D. TEACHER REFLECTION

Which learners explained denaturation as a permanent change in shape rather than 'the enzyme dies'? Did every group control variables and record 'not cleared' correctly? Which data points did learners use most confidently (rate 1.0 at 37 C, 5 times faster than at 20 C)? Did the CER statements include numbers? Which misconceptions (hotter is always faster, cold kills enzymes) remain, and how will they be addressed in Lesson 6? How well did the DQB and the class model show the progress from Lessons 1 to 5? Note any safety issues with the hot water baths and Benedict's test.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Chewed ugali tastes sweeter. In the investigation, starch with saliva amylase cleared fastest at 37 C (60 s, rate 1.0) and at pH 6.8
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	(60 s, rate 1.0). At 20 C it took 300 s (rate 0.2). At 0, 60 and 80 C and at pH 2 and 12, starch was not cleared in 600 s (rate 0). The Benedict's test showed a reducing sugar after amylase acted.
What did I learn?	Enzymes are protein catalysts made of amino acids. Amylase breaks starch into maltose, a reducing sugar. Enzymes work best at an optimum temperature (about 37 C) and pH (near neutral). Extreme heat or pH denatures an enzyme permanently. Each enzyme is specific to its substrate.
How does this explain the phenomenon?	Ugali tastes sweet when chewed because saliva amylase, a protein enzyme, breaks starch into maltose at body temperature. The reaction at 37 C is 5 times faster than at 20 C (1.0 / 0.2). Without enzymes working at their optimum, the starch in ugali could not be turned into glucose quickly enough to fuel a runner like Kipchoge from Iten.

LESSON 6 (80 minutes (double lesson)): Which meal best supports the athlete?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To use evidence from Table 4 and Table 1 to compare three Kenyan athlete meals, argue which one best supports an endurance athlete, and give the final explanation of how simple foods such as ugali, githeri and beans provide the chemicals of life that fuel Kipchoge-type performance.
Knowledge	Learners will explain that a plate of ugali, beans, sukuma wiki and milk is not plain: it contains carbohydrates (starch in maize, supplying and storing energy as glucose and glycogen), proteins (beans, milk, omena; amino acids for growth and repair, and enzymes) and lipids (milk fat, groundnuts, cooking oil; stored energy and insulation). They will use $\text{Energy} = 4 \times \text{C} + 4 \times \text{P} + 9 \times \text{F}$ to compare Meals A, B and C and explain that enzymes such as amylase, working best at about 37 C and near neutral pH, break big food molecules into small ones the body can absorb.
Skills	Analysing and interpreting data (NGSS SEP 4): calculating and comparing energy and energy shares. Engaging in argument from evidence (SEP 7): writing a Claim-Evidence-Reasoning statement. Constructing explanations (SEP 6): giving the final explanation of the phenomenon. Developing and using models (SEP 2): comparing the Lesson 1 model with the final model. Communication and collaboration in groups of 4.
Attitudes	Appreciate that locally available Kenyan foods supply the chemicals of life; respect different opinions during the argument; value evidence over opinion; confidence that they did science, not just learned about it.
Key Inquiry Question	1. What are the chemicals of life? 2. How do chemicals of life contribute to the functioning of living organisms?
Purpose in Storyline	Lesson 6 is the synthesis lesson. Lessons 1 to 5 gave the parts: the three groups of chemicals named (L1), carbohydrate structure and the iodine and Benedict's tests (L2), proteins as chains of amino acids with the Biuret test (L3), lipids with the emulsion and brown paper tests and 9 kcal per g (L4), and enzymes as protein catalysts with optimum temperature and pH (L5). Now learners put the parts together to answer the Driving Question, finalise the single class model and close out the DQB.
Safety Notes	This is a data and discussion lesson, so no chemical tests are carried out. If any food samples are shown, no tasting of anything that has been in the laboratory. Keep walkways clear during gallery walks. Be sensitive to learners who may not have regular meals at home: talk about 'an athlete's meal', not 'your meal'.

B. LESSON OVERVIEW

Learners return to the opening puzzle: how can plain foods like ugali and beans fuel Eliud Kipchoge's 2:01:09 marathon (7,269 s, 42.195 km, about 172.3 s per km)? Mama Wanjiru, a canteen cook in Eldoret, must choose a meal for a group of young runners from Iten. Groups calculate and compare three meals using Table 4 and Table 1 data (Meal A: ugali 300 g + beans 150 g + sukuma wiki 100 g + milk 250 g; Meal B: githeri 400 g + sukuma wiki 100 g; Meal C: mandazi 200 g + nyama choma 100 g). They write a Claim-Evidence-Reasoning argument citing numbers, relate the meal to the marathon cost of about 2,200 kcal and glycogen store of about 2,000 kcal, compare the Lesson 1 model with the final model, answer every question on the Driving Question Board, and write the Final Explanation. The lesson ends with a celebration: 'You didn't just learn ABOUT science, you DID science!'

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Introduction to carbohydrates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Enzymes in the Digestive System (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners meet Mama Wanjiru's problem: 'Three young runners from Iten will train hard tomorrow. I can cook Meal A (ugali, beans, sukuma wiki and milk), Meal B (githeri and sukuma wiki) or Meal C (mandazi and nyama choma). Which one is best?' Individually they write a prediction in 3 minutes: which meal they think best supports an endurance athlete and one reason, without calculating. They also write which of the three chemicals of life they expect each meal to be richest in. They then do a 'stand on your choice' vote at the three corners of the room labelled A, B and C. Reasons are linked back to Kipchoge: his marathon costs about 2,200 kcal and his stored glycogen is about 2,000 kcal.	Display the three meal descriptions (not the numbers yet). Say: 'Mama Wanjiru needs your advice. Before we see any numbers, what do you think, and why? Write your prediction. There is no wrong answer yet.' Give 3 minutes of silent writing. Then say: 'Move to the corner of the meal you chose.' Cold-call 2 learners from each corner: 'Tell us your reason in one sentence.' WAIT TIME 10-15 seconds before each call. Record the vote count on the board. Say: 'Today we will not settle this by opinion. We will settle it with evidence from our data.' Do not tell learners which meal is best.	Commitment to a prediction (Predict-Observe-Explain). Writing and standing in a corner makes each learner's initial idea public and visible, so that later evidence can be compared with it and learners notice how their thinking changes.	Check: (1) every learner has a written prediction with a reason; (2) reasons mention food types or chemicals (for example 'ugali has starch for energy') rather than just 'it tastes good' or 'meat is strong food'; (3) count which misconceptions appear (for example 'meat or fried foods give the most energy', 'mandazi is bad because it is sweet'). Note these on a class tally for revisiting in the Explain phase.
Observe Phase  VIDEO: Food and energy in organisms Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzymes in the Digestive System Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	In groups of 4, learners receive a data card with Table 4 values for the three meals (carbohydrate, protein, fat in g and energy in kcal) and the Table 1 values per 100 g. Task 1 (verify): each learner calculates the energy of one meal using $\text{Energy} = 4 \times \text{C} + 4 \times \text{P} + 9 \times \text{F}$ and checks it against the card: Meal A = $505 + 121 + 137.7 = 763.7$ kcal (using carbohydrate 126.25 g, protein 30.25 g, fat 15.3 g); Meal B = $308 + 106 + 63 = 477$ kcal; Meal C = $360 + 156 + 405 = 921$ kcal. Task 2 (percentage shares): groups work out or check the energy shares: Meal A carbohydrate 66%, protein 16%, fat 18%; Meal B 65%, 22%, 13%; Meal C 39%, 17%, 44%. Task 3 (rank): groups complete a ranking table: energy $\text{C} > \text{A} > \text{B}$; carbohydrate $\text{A} > \text{C} > \text{B}$; protein C	Circulate with a checklist. Say: 'Show me the three parts of your energy sum: carbohydrate, protein, fat. Which factor do you multiply fat by, and why is it bigger?' (Atwater factors 4, 4, 9 kcal per g, from Lesson 4.) Prompt groups that finish early: 'Meal C has the most energy. Does that make it the best for a runner? What else do you need to look at?' Do not give the answer. Pause the class once for a calculation check: 'Hold up your total for Meal B.' (477 kcal.) Cold-call 3 groups to read out their ranking rows; WAIT TIME 10-15 seconds before calling. Ask: 'Where did the carbohydrate in Meal A come from?' Expect: ugali and beans, mainly starch from maize, which they tested in Lesson 2.	Data analysis with a structured ranking table (Analysing and Interpreting Data). Splitting the data into energy, nutrient amounts and energy shares forces learners to see that 'most energy' and 'best for the purpose' are different questions, which prepares them to argue from criteria.	Observable indicators: learners apply $\text{Energy} = 4 \times \text{C} + 4 \times \text{P} + 9 \times \text{F}$ correctly with units (g, kcal); at least 3 of 4 groups obtain 763.7, 477 and 921 kcal; groups can point to a row in the table when asked for evidence; learners notice that Meal C has the highest fat share (44%) and Meal B the lowest (13%). Look for the error of multiplying all three nutrients by 4, and correct it by asking 'Which chemical stores the most energy per g?'

	> A > B; fat C > A > B; fat share highest in C (44%) and lowest in B (13%). Task 4 (link to Kipchoge): groups compare each meal's energy with the marathon cost of about 2,200 kcal and the glycogen store of about 2,000 kcal, noting that no single meal supplies a whole marathon, so athletes eat several meals and rely on stored glycogen. They also use Table 1 to trace where each meal's carbohydrate comes from (for example ugali 26 g per 100 g, beans 21 g per 100 g, mandazi 45 g per 100 g).			
Explain Phase  VIDEO: Introduction to carbohydrates Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Enzymes in the Digestive System (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners write a Claim-Evidence-Reasoning (CER) argument individually, then compare in groups. Claim: which meal best supports an endurance athlete. Evidence: at least three numbers from Table 4 (for example Meal A: carbohydrate 126.25 g, protein 30.25 g, fat 18% of energy, 763.7 kcal). Reasoning: link the numbers to the roles of the chemicals of life. Carbohydrate (starch in ugali and maize) is broken down by enzymes such as amylase (best at about 37 C and near neutral pH) to glucose, which supplies energy and is stored as glycogen (about 2,000 kcal in the body), so the highest carbohydrate meal refills the store the marathon uses. Protein (beans, milk) gives amino acids for growth and repair of muscle and for making enzymes, and Meal A gives 30.25 g, which is adequate. Fat stores a lot of energy (9 kcal per g) but a meal with a very high fat share (Meal C, 44%) gives less carbohydrate than Meal A (90 g compared with 126.25 g). The strongest argument: Meal A best supports	Model one weak and one strong CER on the board: Weak: 'Meal A is best because it is traditional.' Ask: 'What is missing?' WAIT TIME 10-15 seconds. Say: 'A claim needs numbers and a reason that uses what we learned about carbohydrates, proteins, lipids and enzymes.' Provide sentence frames: 'I claim that Meal ___ best supports an endurance athlete because it gives ___ g of carbohydrate, which ___.' Hold a structured academic argument: assign 2 groups to defend A, 1 to defend B and 1 to defend C, then ask learners to switch sides for 2 minutes. Cold-call 4 learners to give one piece of evidence each, and 2 learners to give a reasoning link. Say: 'Do not call a meal best unless you cite its Table 4 numbers.' Revisit the Predict vote: 'Who changed their mind? What evidence changed it?' Address misconceptions from the tally (for example that 'more energy' always means 'best', or that mandazi is useless: it has 45 g carbohydrate per 100 g, but also 15 g of fat). Do not re-teach tests or enzyme data;	Claim-Evidence-Reasoning with role-switching argumentation (Engaging in Argument from Evidence). Defending and then attacking a position makes learners test whether their evidence really supports the claim, and the Final Explanation synthesises Lessons 1 to 5 into one causal account.	Use a 4-point CER rubric: (1) a clear claim; (2) at least 3 correct numbers from Table 4 (for example 126.25 g, 30.25 g, 18%); (3) reasoning that links carbohydrate to glucose and glycogen, protein to amino acids for growth and repair, lipids to stored energy, and enzymes to breaking down food; (4) a rebuttal that deals with Meal C. Success looks like at least 80% of learners scoring 3 or more. Final Explanation indicators: mentions carbohydrates, proteins and lipids with a role for each, includes enzymes (amylase, 37 C, neutral pH), and links back to the marathon energy of about 2,200 kcal and glycogen of about 2,000 kcal.

	an endurance athlete. Groups then take on the counter-argument: 'Meal C has the most energy (921 kcal) and the most protein (39 g), so why not choose it?' Learners respond with evidence: Meal C has less carbohydrate than Meal A and 44% of its energy comes from fat, while Meal A gives 66% of its energy as carbohydrate. They also discuss why Meal B (githeri and sukuma wiki) is a good, affordable choice with 65% carbohydrate but less energy (477 kcal) than Meal A, so would need larger portions. Finally, learners write their Final Explanation of the phenomenon (3 to 5 sentences) answering: how do simple foods supply the energy, repair and body-building chemicals for a 2:01:09 marathon?	refer back: 'In Lesson 5 we saw amylase work fastest at 37 C and near neutral pH, with the rate 5 times faster than at 20 C.' For the Final Explanation say: 'Start with the plain plate, and end with the marathon. Name all three chemicals of life and one role for each, and include enzymes.' Collect 3 Final Explanations to read aloud.		
Driving Question Board (DQB) Creation  VIDEO: Food and energy in organisms Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Enzymes in the Digestive System Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	The class returns to the Driving Question Board started in Lesson 1 and updated in every lesson. In groups, learners take 3 to 4 question cards each and answer them using evidence from Lessons 1 to 6, writing the answer and the lesson where the evidence came from on a sticky note. Example questions and answers: 'Why does a plain plate of ugali and beans give so much energy?' (starch in maize is a carbohydrate, broken down by amylase to glucose and stored as glycogen). 'How do muscles recover and grow?' (proteins from beans, milk and omena supply amino acids for growth and repair). 'Why do athletes need some fat?' (lipids store a lot of energy, 9 kcal per g, and insulate). 'Why does chewed ugali taste sweet?' (saliva amylase breaks starch to maltose, a reducing sugar). Learners mark each question as 'answered',	Say: 'Each question on our board has been waiting for us. Pick one, find the evidence, and show where it came from.' Give 8 minutes. Circulate and ask: 'Which lesson gave you that evidence?' and 'Is your answer complete or only partly answered?' Cold-call 4 groups to read one answered question, with WAIT TIME 10-15 seconds. Praise the use of evidence from lessons and data. Move any partly answered card to a 'next steps' column and name honestly what Lesson 1 only named but did not teach (water, vitamins, minerals and nucleic acids). Ask: 'Which question was hardest to answer, and why?'	Driving Question Board closure (revisiting learner-generated questions). Learners see their own questions answered with their own evidence, which builds ownership and shows that science is a cycle where answers raise new questions.	Check that each answer sticky note cites specific evidence (a data value, test result or lesson). Indicators: at least 80% of original questions are marked answered or partly answered; answers use correct vocabulary (amino acids, fatty acids and glycerol, starch, glucose, enzyme, optimum); at least 5 new questions are added. Note vague answers (for example 'it gives energy') and ask 'Which chemical, and how?'

	'partly answered' or 'new question'. They add at least one new question for further study (for example about water, vitamins, minerals, or what happens to glucose inside cells).			
Model Building Phase  VIDEO: Dehydration synthesis or a condensation reaction <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Enzymes in the Digestive System (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners finalise the single class model on chart paper. First they display their individual Lesson 1 initial models next to the class model. In a gallery walk, they compare the Lesson 1 model with the final model using the prompts: 'What did we add?', 'What did we change?', 'What did we keep?'. The final model must show: the food (ugali, beans, sukuma wiki, milk); the chemicals of life in them (carbohydrates as starch, proteins as chains of amino acids, lipids as fatty acids and glycerol); enzymes (including amylase) breaking down large molecules into small ones (starch to maltose to glucose, protein to amino acids) at about 37 C and near neutral pH; the small molecules absorbed into the body; energy and storage (glucose to energy, glycogen store of about 2,000 kcal), growth and repair of muscle by amino acids, and fat as stored energy and insulation; and the link to Kipchoge's 2:01:09 marathon (about 2,200 kcal). Each group adds one labelled arrow with a data note (for example 'Meal A: 126.25 g carbohydrate'). Learners then complete a 'Then and Now' reflection: 'In Lesson 1 I thought _____. Now I know _____ because _____. ' The lesson ends with the Final Celebration: learners applaud the class and the teacher closes with, 'You didn't just learn ABOUT science, you DID science!'	Say: 'This is our one class model. Today it becomes our final answer to the Driving Question.' Hold up the Lesson 1 model beside the final one: 'Look at the two. What is different?' WAIT TIME 10-15 seconds. Cold-call 4 learners to name one thing added since Lesson 1. Push for mechanisms: 'You drew an arrow from food to energy. What happens along that arrow? Which enzyme? Which small molecule?' Check that the model does not show that fat is bad or that meat is the only strong food. Use a checklist on the board for the model: food, three chemicals, enzymes, absorption, energy and storage, growth and repair, marathon link. Invite 3 volunteers to explain the model aloud to Mama Wanjiru's young runners in simple language. Close by reading the Driving Question aloud and asking learners to say the answer in chorus in one sentence. Run the Final Celebration checklist: 'Do you feel accomplished? Do you value the process? Are you confident?' Ask for a show of hands, then give the closing line.	Cumulative model revision with Then-and-Now comparison (Developing and Using Models). Placing the Lesson 1 model beside the final model makes learning visible: learners see their own growth from a 'plain plate' to a mechanism-based explanation.	Observable indicators: the final model includes all items on the checklist (food, carbohydrates, proteins, lipids, enzymes, small molecules, energy and storage, growth and repair, marathon link); labels use correct terms (amino acids, fatty acids and glycerol, glucose, glycogen, amylase, denatured if shown); at least 3 learners can explain the model without notes; Then-and-Now statements show a real change in thinking, with at least one piece of evidence. Celebration indicators: learners express pride and confidence in using evidence.

D. TEACHER REFLECTION

After the lesson, consider: Did learners support their claim with at least three numbers from Table 4, or did they rely on opinion and tradition? Which misconceptions from the Predict tally remained (for example 'most energy means best', or 'fat or mandazi is always bad')? Did learners link carbohydrate to glucose and glycogen, protein to amino acids, lipids to stored energy and enzymes to digestion in their Final Explanations? Compare the Lesson 1 models with the final model: which ideas changed most? Which DQB questions remain partly answered, and how might they be taken forward (for example water, vitamins, minerals and nucleic acids)? Were all learners, including quieter ones, cold-called and given WAIT TIME? What would you change about the Meal A, B and C comparison for next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We compared Meals A, B and C using Table 4: Meal A gave 763.7 kcal with 126.25 g carbohydrate, 30.25 g protein and 18% of energy from fat; Meal B gave 477 kcal; Meal C gave 921 kcal but 44% of its energy came from fat and only 90 g carbohydrate. We compared Kipchoge's marathon cost of about 2,200 kcal with his glycogen store of about 2,000 kcal, and we compared our Lesson 1 model with our final model.
What did I learn?	Ugali, beans, sukuma wiki and milk contain all three chemicals of life: carbohydrates (starch) for energy, proteins (amino acids) for growth and repair and for making enzymes, and lipids for stored energy and insulation. Enzymes such as amylase work best at about 37 C and near neutral pH and break big food molecules into small ones the body can absorb. $\text{Energy} = 4 \times \text{C} + 4 \times \text{P} + 9 \times \text{F}$ lets us compare meals with numbers.
How does this explain the phenomenon?	A plain plate of ugali and beans is not plain: it supplies carbohydrates that enzymes break down to glucose for energy and glycogen storage, proteins that supply amino acids for repair and growth, and some lipids for stored energy. Meal A (ugali, beans, sukuma wiki and milk) best supports an endurance athlete because it gives the most carbohydrate (126.25 g), adequate protein (30.25 g) and moderate fat (18% of energy). This is how simple Kenyan foods can fuel a 2:01:09 marathon like Eliud Kipchoge's.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.